

AI ASSISTANT CODING

ASSIGNMENT-5.3

Lab 5: Ethical Foundations – Responsible AI Coding Practices

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Batch: 20

Task 1: Privacy and Data Security in AI-Generated Code

To analyze AI-generated login code for privacy and security risks such as hardcoded credentials and plain-text password handling, and to revise it for better security practices.

Prompt: # Create a simple login system in Python using username and password.

Code:

```
ass5.3.py > ...
1  #create a simple login system in python using username and password
2  username = "admin"
3  password = "password123"
4  u=input("Enter username: ")
5  p=input("Enter password: ")
6  if u==username and p==password:
7      print("Login successful!")
8  else:
9      print("Login failed. Incorrect username or password.")
10 import getpass
11 stored_username = "admin"
12 stored_password_hash="hashed_password_value" # Replace with actual hashed password
13 username_input = input("Enter username: ")
14 password_input = getpass.getpass("Enter password: ")
15 if username==stored_username:
16     print("password verification required")
17 else:
18     print("Invalid username")
```

Result:

```

es\Python312\python.exe' 'c:\Users\manus\.vscode\extensions\ms-python.d
AI Assistance\ass5.3.py'
Enter username: admin
Enter password: password123
Login successful!
Enter username: admin
Enter password:
password verification required
PS C:\Users\manus\OneDrive\Documents\Desktop\AI Assistance>

```

Observation:

The AI-generated login system contains significant security vulnerabilities, as the username and password are hardcoded directly into the source code. The password is stored and compared in plain text without any form of encryption or hashing, which exposes sensitive credentials to potential misuse. Additionally, the authentication logic lacks basic security measures such as input validation and secure password handling, making the code unsuitable for real-world applications.

Task 2: Bias Detection in AI-Generated Decision Systems.

This task focuses on identifying bias in AI-generated decision-making logic. The aim is to analyze whether irrelevant attributes such as gender influence outcomes unfairly.

Prompt: # Create a simple loan approval system

Python.

Code:

```

#Task 2: Bias Detection in ai-generated decision systems
def approve_loan(name,gender,income,credit_score):
    if credit_score<600:
        return "Loan Denied"
    elif income<30000:
        return "Loan Denied"
    else:
        return "Loan Approved"
applicant_name = input("Enter applicant name: ")
applicant_gender = input("Enter applicant gender: ")
applicant_income = int(input("Enter applicant income: "))
applicant_credit_score = int(input("Enter applicant credit score: "))
decision = approve_loan(applicant_name,applicant_gender,applicant_income,applicant_credit_score)
print(f"Loan decision for {applicant_name}: {decision}")

```

Result:

```
Enter applicant name: admin
Enter applicant gender: female
Enter applicant income: 50000
Enter applicant credit score: 2000
Loan decision for admin: Loan Approved
```

Observation:

The AI-generated loan approval logic demonstrates clear bias by using gender as a determining factor in the approval process. Since gender is unrelated to an applicant's financial credibility, its inclusion results in unfair treatment of applicants. Such logic can lead to discriminatory outcomes and violates ethical principles of fairness and equality in decision-making systems.

Task 3: Transparency and Explainability in AI-Generated Code

The purpose of this task is to evaluate the transparency and clarity of AI-generated code by analyzing a recursive binary search implementation and its explanation.

Prompt: # Write a recursive binary search program in Python with comments and explanation.

Code:

```
#Task-3:Transparency and Explainability in ai-generated code
def binary_search(arr,low,high,target):
    if high>=low:
        mid=(high+low)//2
        if arr[mid]==target:
            return mid
        elif arr[mid]>target:
            return binary_search(arr,low,mid-1,target)
        else:
            return binary_search(arr,mid+1,high,target)
    else:
        return -1
arr=[2,3,4,10,40]
target=10
result=binary_search(arr,0,len(arr)-1,target)
if result!=-1:
    print(f"Element is present at index {result}")
else:
    print("Element is not present in array")
```

Result:

```
Document (Desktop) AI Assistant (2024-08-08)
Element is present at index 3
PC (C:\Users\user\AppData\Local\Programs\Python\Python310\Scripts\python.exe)
```

Observation:

The AI-generated recursive binary search program is transparent and easy to understand, as it clearly defines both the base case and the recursive case. The inline comments accurately describe each step of the algorithm, and the overall explanation aligns well with the implemented logic. As a result, the code is suitable for beginner-level learners and meets the requirements of explainability and clarity.

Task 4: Ethical Evaluation of AI-Based Scoring Systems

This task evaluates whether AI-generated scoring systems use ethical and objective criteria when assessing job applicants.

Prompt: #Create a job applicant scoring system based on skills, experience, and education.

Code:

```
#task 4:
def score_applicant(name,gender,skills,experiance,education):
    score=skills*2+experiance*3
    if education=="bachelors":
        score+=2
    if gender=="female":
        score+=2
    return score
score1=score_applicant("Alice","female",5,3,"bachelors")
score2=score_applicant("Bob","male",5,3,"bachelors")
print(f"Alice's score: {score1}")
print(f"Bob's score: {score2}")
```

Result:

```
Alice's score: 23
Bob's score: 21
```

Observation

The applicant scoring system generated by the AI introduces ethical concerns by assigning additional score based on gender. Since gender and name are unrelated to professional capability or job performance, their inclusion results in biased and unfair scoring. Ethical scoring systems should rely solely on objective and job-relevant attributes such as skills, experience, and education

Task 5: Inclusiveness and Ethical Variable Design.

The goal of this task is to identify non-inclusive variable usage and assumptions related to gender or identity in AI-generated code.

Prompt: #Generate Python code to process employee details.

Code:

```
#Task 5:
# Inclusiveness and Ethical Variable Design
# generate python code to process employ details
#
def process_employee_details(name,gender):
    if gender=="female":
        print("Assign to HR for further processing")
    else:
        print("Assign to General department for further processing")
process_employee_details("Alice","female")
process_employee_details("Bob","male")
process_employee_details("Charlie","non-binary")
```

Result:

```
Assign to HR for further processing
Assign to General department for further processing
Assign to General department for further processing
PS C:\Users\manus\OneDrive\Documents\Desktop\AI Assistance>
```

Observation:

The AI-generated code makes assumptions about employee roles based on gender, which reinforces stereotypes and excludes non-binary identities. Assigning departments purely on gender ignores individual skills, interests, and qualifications. Such logic is not inclusive and highlights the need for ethical variable design that avoids identity-based assumptions.

